

OpenCV: An Excellent Tool for Computer Vision

Computer vision imparts human intelligence and instincts to a computer. This field of computer science works on enabling computers to see, identify and process images the same way the human eye does. OpenCV is a great tool to accelerate computer vision in commercial products.

Let us say you and your family went on a vacation and you uploaded a few pictures on Facebook. However, since it takes time to tag the names in each picture, Facebook is intelligent enough to do that for you. How do you think this auto-tag feature works? Well, it works through computer vision.

What is OpenCV?

OpenCV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in commercial products. This library has more than 2500 algorithms used to detect and recognise human faces, identify images, track moving objects, and extract 3D models of objects.

Installation of OpenCV

To install OpenCV for Python, use the following code in the terminal:

```
$ python3 -m pip install opencv-python
$ python3 -m pip install opencv-contrib-python
```

How does a computer read an image?

One look at Figure 2, and we can see that it is a picture of the Times Square in New York. But computers cannot

analyse that, since they don't have any intelligence.

For any image, there are three primary colours — red, green and blue. A matrix is formed for every primary colour, and later, these matrices combine to provide a pixel value for the individual R, G, and B colours. Each element of these matrices provides data pertaining to the intensity of brightness of the pixel. It reads any image as a range of values between 0 and 255.

How are videos and images captured through a camera?

```
import cv2
cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    cv2.imshow("Capturing", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

As seen in the above piece of code, we need to import the OpenCV library, `cv2`. `VideoCapture()` triggers the camera, and `cv2.imshow` shows what the camera is capturing by opening a new window. `cv2.waitKey` makes the window static until the user presses a key.

Basic functions of OpenCV

To load images using OpenCV and

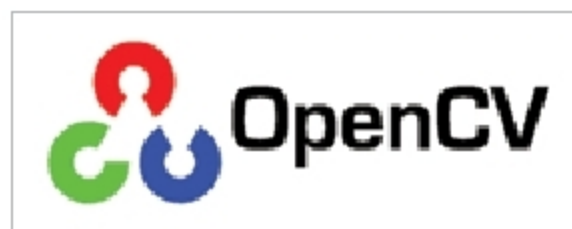


Figure 1: Logo of OpenCV



Figure 2: New York's Times Square

